

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Dry Run Evaluation

Course Code:	CSA17	Course Title:	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	Assessment:	Assessment Tool 2 – Dry Run Evaluation
Weightage:	35% of CIA	Duration / Marks:	60 mins / 100 marks
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	B.tech-IT	Date:	10/10/2026

Code of Conduct

I Dinesh R-192421117 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: R Dinesh

Instructions

- Read each question carefully; answer all four questions (25 marks each, 100 in total).
- Show every dry-run step: the initial queue, stack, priority queue or belief state, each iteration, your assumptions and a clear final answer, and finish with the interpretation asked for.
- Follow the conventions stated in each question (alphabetical expansion, explored set, goal test on removal from the frontier, tie-breaking) and present tables neatly.
- Marks follow the Dry Run Evaluation rubric: understanding of the search problem, state expansion and dry run execution, application of the search algorithm, accuracy of the solution path, and presentation and logical reasoning.
- Upload in GitHub and GCR (question paper with Reg. No., Name, answers, rubrics and evaluation page at the end). Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Four approaches to AI, Turing Test, eras and milestones	Q2
2. Intelligent Agents: Agents and Environments	Agent, percepts, actions, agent function and program, PEAS	Q1
3. The Concept of Rationality	Performance measure, percept sequence, rational vs omniscient	Q1
4. The Nature of Environments	Task-environment properties and their classification	Q1
5. The Structure of Agents	Simple reflex and model-based reflex agent programs	Q1
6. Problem-Solving Agents	Goal formulation, problem formulation, search, execution	Q2
7. Problem Formulation	States, initial state, actions, goal test, path cost; belief states	Q2, Q4
8. Searching for Solutions	Search graph, frontier, explored set, node expansion	Q3
9. Uninformed Search Strategies	Dry runs of BFS, DFS and uniform-cost search	Q3

10. Search with Partial Information

Sensorless problems, belief states,
percept update

Q4

Annexure A – Dry Run Evaluation

Question 1: Agents, Environments and Rationality – Dry Run of Agent Programs

Topic: Intelligent agents, rationality, nature of environments, structure of agents | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives [its location, status of its own square] and performs one action: Suck, Right, Left or NoOp. The agent starts in square A, both squares are initially Dirty, and dirt never reappears.

Performance measure over T = 6 time steps: +10 for every square cleaned (a Suck on a Dirty square) and –1 for every Left or Right move.

Agent X (simple reflex): if the current square is Dirty, Suck; otherwise move to the other square.

Agent Y (model-based reflex): keeps an internal state of the squares it knows to be clean. If the current square is Dirty, Suck; otherwise, if both squares are known clean, NoOp; otherwise move to the other square.

a) PEAS description

PEAS component	Answer
Performance measure	+10 for each square cleaned; –1 for each Left/Right move over 6 steps.
Environment	Two squares A and B; each may be Dirty or Clean; dirt does not reappear.
Actuators	Suck, Left, Right, NoOp.
Sensors	Perceives current location and status (Dirty/Clean) of the current square.

(b) Task-environment classification

Property	Classification	Reason
Observability	Partially observable	The agent sees only the current square, not the other square.
Number of agents	Single-agent	Only the vacuum agent acts.
Determinism	Deterministic	Actions have predictable effects.
Episodic / Sequential	Sequential	Current actions affect later states and choices.
Static / Dynamic	Static	Dirt and world do not change independently during an action.
Discrete / Continuous	Discrete	Squares, dirt states, actions and time steps are discrete.

(c) Dry-run of both agents

Step	X percept	X action	Y percept	Y action	Y known-clean squares after step
1	[A, Dirty]	Suck	[A, Dirty]	Suck	{A}
2	[A, Clean]	Right	[A, Clean]	Right	{A}
3	[B, Dirty]	Suck	[B, Dirty]	Suck	{A, B}
4	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}
5	[A, Clean]	Right	[B, Clean]	NoOp	{A, B}
6	[B, Clean]	Left	[B, Clean]	NoOp	{A, B}

Assumption: after each movement, the next step begins in the square reached. Agent Y stays in B after it knows both squares are clean.

(d) Total performance over 6 steps

Agent X cleans A and B: $2 \times 10 = 20$ points. It makes four moves (steps 2, 4, 5 and 6): $4 \times (-1) = -4$. Total X = $20 - 4 = 16$.

Agent Y cleans A and B: $2 \times 10 = 20$ points. It makes one move (step 2): -1 . It uses NoOp after both squares are known clean. Total Y = $20 - 1 = 19$.

(e) Interpretation

Agent Y is more rational for this performance measure because it remembers which squares are clean and stops moving once both are clean, avoiding unnecessary penalties. At step 4, both agents receive [B, Clean], but Agent X has no internal state, so its simple-reflex rule says to move to the other square. Agent Y remembers that A and B are both clean, so it chooses NoOp. A rational agent selects actions that maximize expected performance using its percepts and internal information. An omniscient agent would know the actual state and outcome perfectly; a rational agent does not need perfect knowledge, only the best decision based on available information.

Question 2: Foundations and History of AI; Problem-Solving Agent and Problem Formulation

Topic: Foundations and history of AI, problem-solving agents, problem formulation | 15 minutes

Part A. Complete the table by classifying each system or idea under one of the four approaches to AI (thinking humanly, acting humanly, thinking rationally, acting rationally) and naming the era or milestone of AI history to which it belongs.

Part A – (a) Four approaches and AI milestones

System / idea	Approach to AI	Era / milestone
Turing Test proposed (1950)	Acting humanly	1950: Alan Turing proposes the imitation game/Turing Test.
General Problem Solver (Newell and Simon, 1957)	Thinking humanly	1950s: early symbolic AI and cognitive problem-solving models.
Aristotle's syllogisms and formal logic	Thinking rationally	Ancient foundations of logic; formal rules for valid reasoning.
An agent that does the right thing to maximise expected performance	Acting rationally	Modern rational-agent approach to AI.

Part B – Two-jug problem

Question details: Jug X holds 4 litres and jug Y holds 3 litres. Both start empty. The actions are fill, empty, and pour until the source is empty or the target is full. Goal: exactly 2 litres in X.

(b) Problem formulation

Element	Answer
State representation	(x, y) , where x is the amount in X (0–4) and y is the amount in Y (0–3).
Initial state	$(0, 0)$.
Actions	Fill X, fill Y, empty X, empty Y, pour $X \rightarrow Y$, pour $Y \rightarrow X$.
Transition model	Apply the chosen action; pouring continues until the source is empty or the target is full.
Goal test	$x = 2$ litres.
Path cost	1 per action; total cost is the number of actions.

(c) Successor states

From $(0, 0)$, excluding actions that leave the state unchanged: $\{(4, 0), (0, 3)\}$.

From $(4, 0)$, excluding actions that leave the state unchanged: $\{(0, 0), (4, 3), (1, 3)\}$. The state $(1, 3)$ results from pouring X into Y until Y is full.

(d) State-space size and shortest solution

There are 5 possible amounts in X (0, 1, 2, 3, 4) and 4 possible amounts in Y (0, 1, 2, 3). Therefore, the full state-space upper bound is $5 \times 4 = 20$ states.

Step	Action	State (X, Y)
0	Start	$(0, 0)$
1	Fill Y	$(0, 3)$
2	Pour $Y \rightarrow X$	$(3, 0)$
3	Fill Y	$(3, 3)$
4	Pour $Y \rightarrow X$ until X is full	$(4, 2)$
5	Empty X	$(0, 2)$
6	Pour $Y \rightarrow X$	$(2, 0)$

Goal reached: X contains exactly 2 litres. Path cost = 6 actions.

(e) Interpretation

The agent first formulates the goal (get exactly 2 litres in X), then formulates the states and legal jug actions. It searches the state space to find the six-action sequence and then executes those actions in order. The plan can be executed without new percepts because the jug capacities and action effects are known and deterministic, assuming the environment behaves as specified.

Question 3: Searching for Solutions – Uninformed Search Strategies *Topic: Searching for solutions, BFS, DFS, uniform-cost search | 15 minutes*

Consider the weighted undirected graph below. The start node is S and the goal node is G. Use graph search with an explored set; expand neighbours alphabetically, test the goal when a node leaves the frontier, and break ties alphabetically. BFS and DFS ignore edge costs while searching, but path costs are measured with them.

Edge	Cost	Edge	Cost
S – A	1	B – D	1
S – B	4	C – G	6
A – C	2	D – G	2
A – D	5		

(a) Breadth-first search (BFS)

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[A, B]	{S}
2	A	[B, C, D]	{S, A}
3	B	[C, D]	{S, A, B}
4	C	[D, G]	{S, A, B, C}
5	D	[G]	{S, A, B, C, D}
6	G	[] (goal)	{S, A, B, C, D, G}

BFS path: $S \rightarrow A \rightarrow C \rightarrow G$. Path cost = $1 + 2 + 6 = 9$. Nodes removed from frontier = 6.

(b) Depth-first search (DFS)

Stack top is shown on the left. Children are pushed in reverse alphabetical order so the alphabetically first child is popped first.

Iteration	Node popped	Stack after pushing children
1	S	[A, B]
2	A	[C, D, B]
3	C	[G, D, B]
4	G	[D, B] (goal)

DFS path: $S \rightarrow A \rightarrow C \rightarrow G$. Path cost = 9. Nodes removed from frontier = 4.

(c) Uniform-cost search (UCS)

Iteration	Node expanded	$g(n)$	Priority queue after expansion (lowest cost first)
1	S	0	[A:1(S), B:4(S)]
2	A	1	[C:3(A), B:4(S), D:6(A)]
3	C	3	[B:4(S), D:6(A), G:9(C)]
4	B	4	[D:5(B), G:9(C)]
5	D	5	[G:7(D)]
6	G	7	[] (goal)

When B is expanded, the path to D improves from cost 6 to cost 5. When D is expanded, the path to G improves from cost 9 to cost 7.

UCS path: $S \rightarrow B \rightarrow D \rightarrow G$. Path cost = $4 + 1 + 2 = 7$. Nodes removed from frontier = 6.

(d) Comparison

Strategy	Path found	Path cost	Nodes removed	Optimal here?
BFS	$S \rightarrow A \rightarrow C \rightarrow G$	9	6	No (not lowest cost)
DFS	$S \rightarrow A \rightarrow C \rightarrow G$	9	4	No
UCS	$S \rightarrow B \rightarrow D \rightarrow G$	7	6	Yes

(e) Interpretation

UCS removes as many nodes as BFS here, but it considers path costs and finds the cheaper path of cost 7. BFS finds a path with fewer edges but ignores edge weights, so its path costs 9. DFS reaches the goal after removing only four nodes in this graph, but it is not generally reliable for finding the cheapest or shallowest solution and may follow a poor path. An explored set prevents repeatedly expanding already visited nodes in an undirected graph. For example, during BFS iteration 3, B has a connection back to S and to D; S is already explored and D is already in the queue, so neither is added again.

Question 4: Search with Partial Information – Belief States

Topic: Problem formulation, search with partial information (sensorless and percept-based belief-state search) | 15 minutes

A vacuum agent lives in a two-square world (Left square, Right square), each square being Clean (C) or Dirty (D). Actions: Left, Right and Suck (Left in the left square and Right in the right square leave the agent where it is; Suck cleans the current square). Physical states are written (Agent, Left square, Right square):

State	Description	State	Description
S1	(L, D, D)	S5	(L, C, D)
S2	(R, D, D)	S6	(R, C, D)
S3	(L, D, C)	S7	(L, C, C)
S4	(R, D, C)	S8	(R, C, C)

(a) Sensorless belief-state problem formulation

Element	Answer
Initial belief state	{S1, S2, S3, S4, S5, S6, S7, S8}
Actions	Left, Right, Suck.
Goal test	Every physical state in the belief state has both squares Clean (belief state is a subset of {S7, S8}).
Path cost	Assume unit cost per action; minimize the number of actions.

(b) Belief state after each sensorless action

Step	Action	Belief state after action
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}
1	Right	{S2, S4, S6, S8}
2	Suck	{S4, S8}
3	Left	{S3, S7}
4	Suck	{S7}

(c) Belief-state prediction and percept update

Step	Action	After action (predicted)	Percept	After percept (updated)
0	Start	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

(d) Number of possible belief states

With 8 physical states, the number of possible subsets (belief states) is $2^8 = 256$, including the empty set. This is harder than searching physical states because the search may need to consider up to 256 belief states rather than only 8 physical states.

(e) Interpretation

A percept removes physical states that do not match what the sensor reports. The first percept [Left, Dirty] reduces the initial belief state from eight states to {S1, S3}. After Suck, the predicted states are {S5, S7}; the [Left, Clean] percept is consistent with both, so the belief state remains {S5, S7}. After Right, the predicted belief state is {S6, S8}; [Right, Dirty] removes S8 and leaves {S6}. In the sensorless case, no percepts are available, so the agent cannot shrink its belief state using observations and must choose a plan that works across all possible states. Such a plan can still be rational if it is selected to achieve the goal under the available information.

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Search Problem	20	Clearly understands the search problem, initial state, goal state, and search strategy.	Understands the problem with minor misunderstandings.	Demonstrates partial understanding of the search problem.	Shows little or no understanding of the search problem.
State Expansion and Dry Run Execution	30	Correctly traces every iteration, expands nodes accurately, and maintains the Queue/Stack/Priority Queue without errors.	Performs most iterations correctly with minor mistakes in state expansion or data structure updates.	Performs only some iterations correctly; several errors in tracing or node expansion.	Unable to correctly perform the dry run or expand states.

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Application of Search Algorithm	20	Applies the selected algorithm (BFS/DFS/UCS) correctly, following all algorithmic rules.	Applies the algorithm correctly with a few procedural errors.	Applies only basic algorithm steps with noticeable mistakes.	Fails to apply the correct search algorithm.
Accuracy of Solution Path	20	Identifies the correct traversal order/solution path and goal state with proper justification.	Solution path is mostly correct with minor errors.	Partially correct solution path; goal may not be reached correctly.	Incorrect or incomplete solution path.
Accuracy of Solution Path	10	Queue/Stack/Priority Queue updates, state diagrams, and explanations are neat, organized, and logically presented.	Presentation is clear with minor organizational issues.	Limited explanation and organization; reasoning is partially clear.	Poor presentation with unclear or missing reasoning.

Q. No. / Criteria	Understanding of Search Problem	State Expansion and Dry Run Execution	Application of Search Algorithm	Accuracy of Solution Path	Accuracy of Solution Path	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
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Date: _____	Date: _____	Date: _____